

Problem 22 Statement:Find the derivative with respect to x of the function:

$$y = e^{\sqrt{x}} \cdot \sqrt{x^2 - 1}$$

ধাপ ১: প্রদত্ত ফাংশনটি উল্লেখ করি (Given Function):

$$y = e^{\sqrt{x}} \cdot \sqrt{x^2 - 1}$$

ধাপ ২: উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating w.r.t. x):

$$\frac{dy}{dx} = \frac{d}{dx} \left[e^{\sqrt{x}} \cdot \sqrt{x^2 - 1} \right]$$

ধাপ ৩: গুণন সূত্র বা Product Rule প্রয়োগ করি $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$:

$$\frac{dy}{dx} = e^{\sqrt{x}} \cdot \frac{d}{dx} (\sqrt{x^2 - 1}) + \sqrt{x^2 - 1} \cdot \frac{d}{dx} (e^{\sqrt{x}})$$

ধাপ ৪: প্রতিটি অংশের ওপর চেইন রুল প্রয়োগ করে অন্তরক নির্ণয় করি (Applying Chain Rule):

$$\frac{d}{dx} (\sqrt{x^2 - 1}) = \frac{1}{2\sqrt{x^2 - 1}} \cdot \frac{d}{dx} (x^2 - 1) = \frac{2x}{2\sqrt{x^2 - 1}} = \frac{x}{\sqrt{x^2 - 1}}$$

$$\frac{d}{dx} (e^{\sqrt{x}}) = e^{\sqrt{x}} \cdot \frac{d}{dx} (\sqrt{x}) = e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} = \frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

ধাপ ৫: অন্তরজের মানসমূহ বসিয়ে $e^{\sqrt{x}}$ কমন নিয়ে সরলীকরণ করি:

$$\frac{dy}{dx} = e^{\sqrt{x}} \cdot \frac{x}{\sqrt{x^2 - 1}} + \sqrt{x^2 - 1} \cdot \frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

$$\frac{dy}{dx} = e^{\sqrt{x}} \left(\frac{x}{\sqrt{x^2 - 1}} + \frac{\sqrt{x^2 - 1}}{2\sqrt{x}} \right)$$

$$\text{Ans : } \frac{dy}{dx} = e^{\sqrt{x}} \left(\frac{x}{\sqrt{x^2 - 1}} + \frac{\sqrt{x^2 - 1}}{2\sqrt{x}} \right)$$

ENGINEERING INSIGHTS & EXAM NOTES

AMIE Section-A Examination Strategy & Rigor Analysis

📌 BOX 1: FORMULATION & ENGINEERING CORE FORMULAS

- **Leibniz Product Rule:** $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$.
- **Exponential Derivative:** $\frac{d}{dx}(e^{g(x)}) = e^{g(x)} \cdot g'(x)$.
- **Square Root Composite:** $\frac{d}{dx}(\sqrt{g(x)}) = \frac{g'(x)}{2\sqrt{g(x)}}$.
- **Alternative Combined Form:** Bringing to a common denominator:

$$\frac{dy}{dx} = e^{\sqrt{x}} \left[\frac{2x\sqrt{x} + (x^2 - 1)}{2\sqrt{x}\sqrt{x^2 - 1}} \right] = \frac{e^{\sqrt{x}}(2x^{3/2} + x^2 - 1)}{2\sqrt{x}(x^2 - 1)}$$

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Forgetting the Inner Derivative of $\sqrt{x}$:** Writing $\frac{d}{dx}(e^{\sqrt{x}}) = e^{\sqrt{x}}$ instead of multiplying by $\frac{1}{2\sqrt{x}}$.
- **Missing Factor of 2 in $\frac{d}{dx}(x^2 - 1)$:** $\frac{d}{dx}(x^2 - 1) = 2x$. The factor 2 in the numerator cancels with the 2 in the denominator $2\sqrt{x^2 - 1}$.
- **Premature Multiplication:** Do not expand the product blindly without factoring out the shared $e^{\sqrt{x}}$ term; factoring keeps the final result clean and fully credited.

🌐 BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Inner Square Roots Domain:** For $\sqrt{x}$, we require $x \geq 0$. For $\sqrt{x^2 - 1}$, we require $x^2 - 1 \geq 0 \implies |x| \geq 1$.
- **Intersection of Domains:** The original function is defined on $[1, \infty)$.
- **Differentiability Restriction:** Because $\sqrt{x}$ and $\sqrt{x^2 - 1}$ appear in denominators of the derivative, $\frac{dy}{dx}$ is strictly valid on $(1, \infty)$. At $x = 1$ and $x = 0$, the derivative diverges to $+\infty$.

💡 BOX 4: LOGARITHMIC DIFFERENTIATION METHOD (ALTERNATIVE)

- Taking natural logarithm on both sides:

$$\ln y = \ln(e^{\sqrt{x}}) + \ln(\sqrt{x^2 - 1}) = \sqrt{x} + \frac{1}{2} \ln(x^2 - 1)$$

- Differentiating implicitly w.r.t. x :

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2\sqrt{x}} + \frac{1}{2} \cdot \frac{2x}{x^2 - 1} = \frac{1}{2\sqrt{x}} + \frac{x}{x^2 - 1}$$

- Multiplying back by $y = e^{\sqrt{x}}\sqrt{x^2 - 1}$:

$$\frac{dy}{dx} = e^{\sqrt{x}}\sqrt{x^2 - 1} \left[\frac{1}{2\sqrt{x}} + \frac{x}{x^2 - 1} \right] = e^{\sqrt{x}} \left(\frac{\sqrt{x^2 - 1}}{2\sqrt{x}} + \frac{x}{\sqrt{x^2 - 1}} \right)$$

- Logarithmic differentiation confirms the exact same result!

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct identification of Product Rule formula $\frac{d}{dx}(uv) = uv' + vu'$.
- **2.0 Marks:** Accurate execution of inner derivatives using Chain Rule for both $e^{\sqrt{x}}$ and $\sqrt{x^2 - 1}$.
- **1.0 Mark:** Factoring out $e^{\sqrt{x}}$ and presenting the finalized neat algebraic expression.